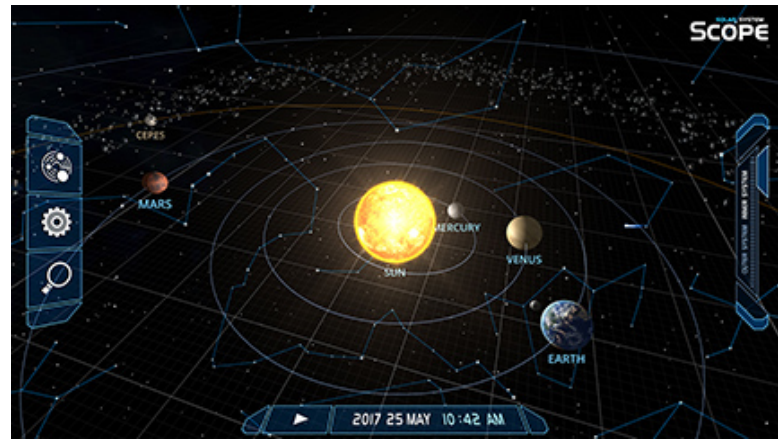


Interactive 1 - Solar System scope

Solar System Scope

This interactive provides a detailed view of the Solar System, in particular the Sun, planets, dwarf planets, moons and comets. There are several variables that can be adjusted such as the point of view, time/date of viewing, size of the planet and the orbit etc.

Instructions to use interactive:



- Click the interactive to start.
- The slider on the right controls zooming - move it up = zoom in, move it down = zoom out. Scrolling with the mouse also controls zooming - scroll up = zoom in, scroll down = zoom out.
- The “**planet**” button () on the left displays the Solar System viewed from the outer space.
- The “**telescope**” button () on the left displays the Solar System viewed from Earth.
- The “**settings**” button () on the left provides options to select for viewing the Solar System.
- The “**search**” button () on the left is used to search for a specific planet, dwarf planet, comet, constellation, or major star.
- Click on the Sun or a planet in the solar system to access quick information about it.
- Double click on the Sun or a planet/dwarf planet/moon to zoom in and access more information about it:
 - Select “**orbit**” to view the object very closely
 - Select “**planet system**” (only available for the planets and moons) to view the planet system it belongs to
 - Select “**structure**” (available to certain objects) to view what it looks like inside (internal structure)
 - Select “**show info**” to access quick information or to view labels of an object’s internal structure
- The “**play**” button on the bottom shows how the Solar System moves over time. Move the slides to increase/decrease the time forwards/backwards.
- Click the date to choose a specific day, month and year to view the Solar System (most useful in “**telescope**” view).
- Click the “**world**” button on the bottom to select the location of the Earth to view the Solar System (most useful in “**telescope**” view).
- Retrograde motion of planets can be observed in the “**telescope**” view by selecting a planet of choice (innermost planets are easiest) and clicking the “**play**” button with the following settings:
 - Tick “keep same day-time”
 - Move the slider to “days”

Activities to complete:

Use the interactive to find information for the following questions:

1. “**Explore**” each of the following planets and use the “**encyclopedia**” to complete the following table.

<u>Planet Name & Picture</u>	<u>Rotation time</u>	<u>Solar Orbit Period</u>	<u>2 interesting facts about the planet</u>
Mercury			
Venus			
Earth			
Mars			
Jupiter			
Saturn			
Uranus			
Neptune			

2. Adjust the settings so the star constellation names is on SHOW.
Click on Earth, and select to view from the “**Night Sky**”.
Name 5 star constellations in the night sky we can see from Earth.
 - i.
 - ii.
 - iii.
 - iv.
 - v.
3. Find and name an example of a **dwarf planet** in our solar system. (make sure this is on SHOW in the setup menu)
4. Find and name an example of a **human spacecraft** in our solar system.
5. At the bottom of the screen there is a “” button. Speed up time and describe how the **orbits** of objects in our solar system change depending on their **distance to the sun**.

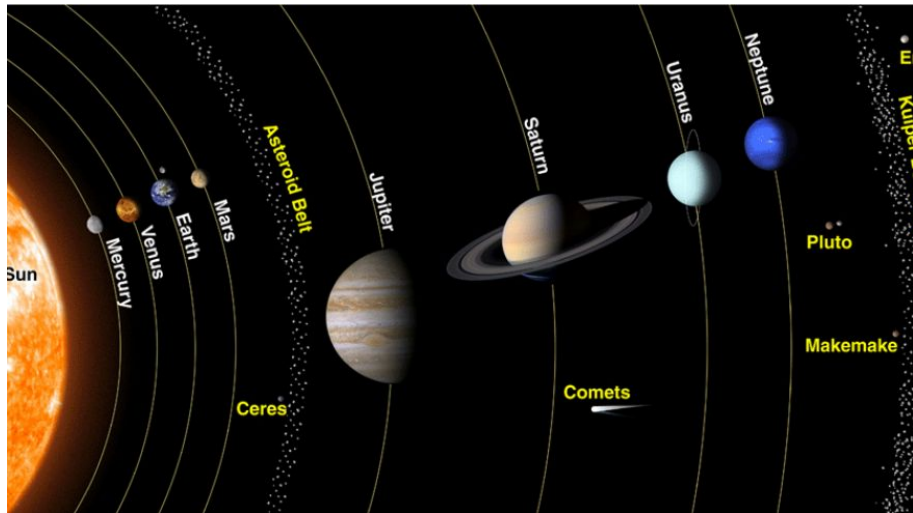
Activity 2: Understanding Models of the Solar System

6. Watch the following video <https://www.youtube.com/watch?v=zR3lgc3Rhfg>

a. **Outline** the **purpose** of the video; why was it made?

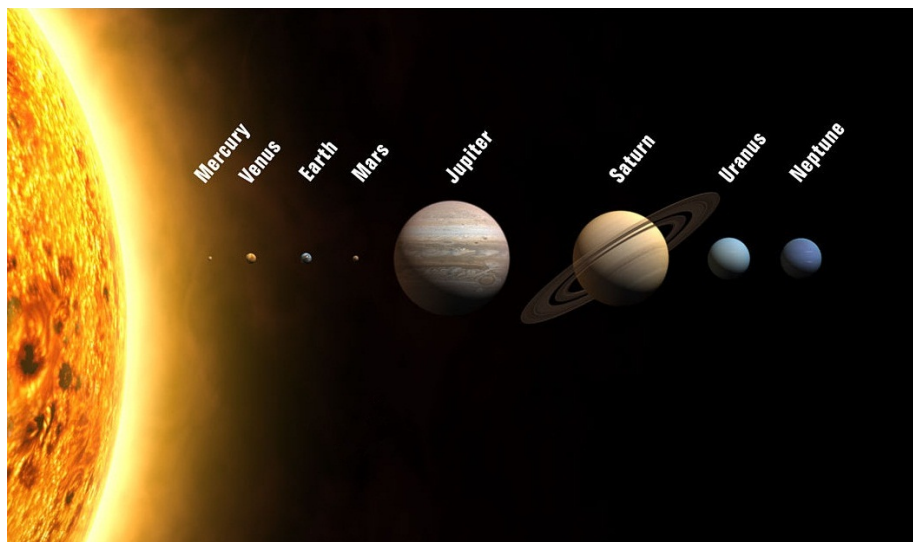
b. **Explain** what they accomplished in the video and the **benefits** of this model of the solar system.

7. Below are 2 different **models** of the solar system that show different aspects of our solar system. Answer the questions in the boxes.



What information does this model show?

- 1.
- 2.
- 3.



What information is this model showing?

- 1.
- 2.
- 3.

Which model do you think is the better model of the solar system? Why?

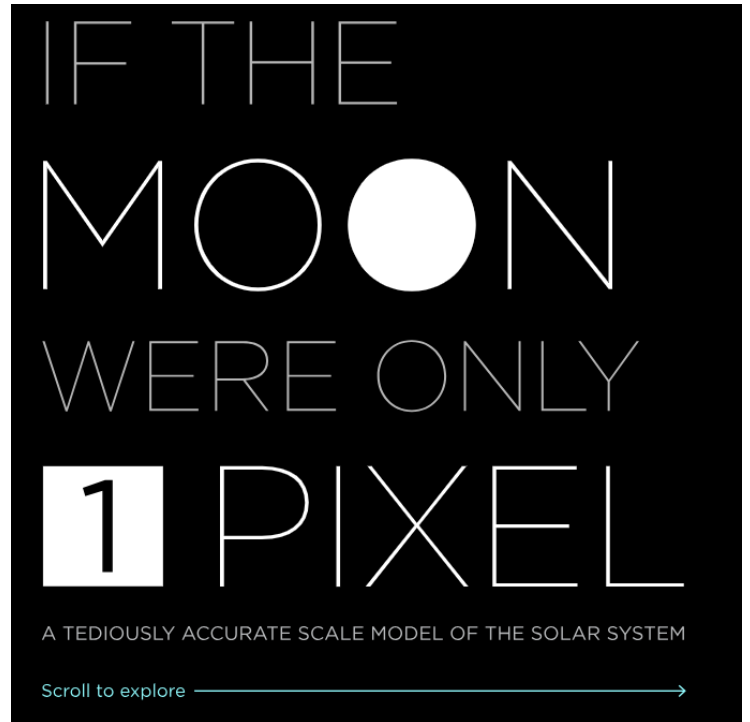
What are some *limitations* of the model you chose?

Explain WHY we use models in Science in our learning (write 2-3 sentences and use an example)

8. Interactive model of the solar system,
[“if the moon were one pixel”](#)

Use the interactive to answer the questions:

- a) What is the benefit of this model of the interactive? Why is it useful?
- b) What are the different measurement scales that are used (at the bottom of the screen
- c) Which unit of measurement is the **LONGEST** out of the options?



- d) Based on your answer for question C, why is it better to use this unit of measurement instead of kilometers?